

TIC 43791794

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 16.267225 d
Epoch (t0) 3463.1432 BTJD
Duration 0.2440 d
Depth 0.047 % (474 ppm)
SNR 23.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

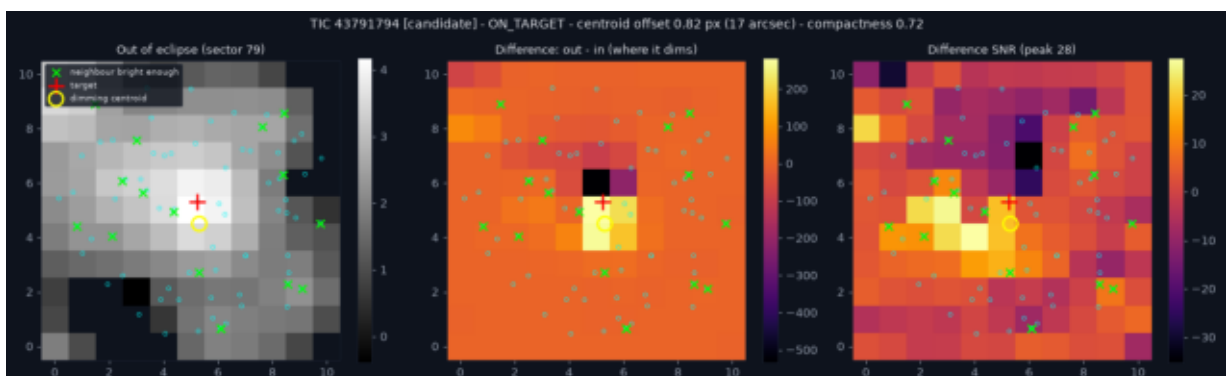
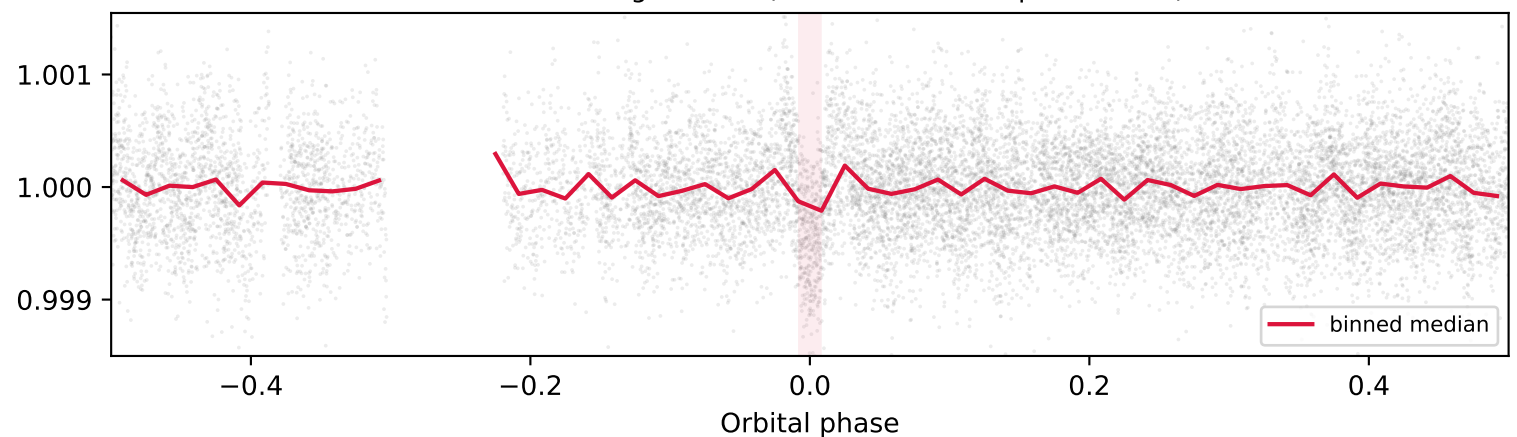
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 15.4 (gate ≥ 7.0)
Odd/even z-score 2.30 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 79 light-curve dip in this sector: 15.7 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 27.66 (gate ≥ 5.0) Compactness 0.72 (gate ≥ 0.50)
Dimming centroid offset 0.82 px (17 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 43791791
1.05 px away, dTmag +5.51 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.